LoRa APRS Tracker V5.2.1 ??? Quick Guide

LoRa APRS Tracker V5.2.1 — Quick Guide

Board: T-Beam S3 Supreme (128×64 OLED) · **Version:** 5.2.1 (base 4.0.1 + APRS screens + FSK + MORSE + WSPR) · **Screens author:** EA7JRS

0. At power-on: choose a mode

Welcome → **SELECT MODE**. **1P** = next, **LP** = accept; after 8 s the last APRS mode starts.

Mode	What it is
APRS LoRa	The usual Tracker (11 pages)
APRS FSK 9600	AX.25 G3RUH at 9600 baud (11 pages; not heard by 1200 baud equipment)
MORSE	A single MORSE screen; never saved and never starts by itself
WSPR beacon	GPS-timed WSPR beacon (single WSPR screen); never saved

To change mode: **MAIN → LP → Change mode** (in MORSE: LP → *Change mode*).

1. One button

Gesture	On the normal screen	With a submenu open
1P (short press)	Next page	Next option
LP (long press)	Opens the page submenu	Runs the highlighted option
2P (double press)	Classic configuration menu	Closes the submenu
3 or 4 presses	WiFi AP menu (as always)	—
5 short or ··· −−− ···	**SOS beacon** (see §4)	—

In the classic menu, 2P returns to the screens. The icon bar at the bottom appears when you change page and hides after 3 s.

2. The 11 pages (APRS modes, in this order)

Page	What it shows	Submenu (LP)
MAIN	Callsign, countdown to next beacon, UTC time, position, locator, last Rx, unread messages	Send beacon · Beacon mode · Next profile · Send SOS · Display off · Config menu · Change mode
MESSAGES	Received and sent messages with status (acknowledged, unacknowledged or NO ACK)	Read / reply · Browse list · Mark all read · Delete all · (in a message) Reply · Quick reply · Delete
LAST APRS	Last station heard: APRS icon, type, distance, bearing, S-meter (`S 8`, `S 9+10`), comment	Next view · Older · Newer · **GO TO**
STATIONS	List of up to 20 stations with distance and age	Browse · Sort · Filter · Clear list
GPS	Coordinates (DD, 6 decimals), locator, altitude, satellites, HDOP, heading	GPS on/off · Coords · Units · SBAS · Constellations
COMPASS	Compass rose and arrow to a station (GO TO)	Target · Next station · Heading · North · Declination · Reset align
ALTIMETER	Altitude, vertical speed (0 when stopped), QNH and altitude profile	Span · Reset profile · QNH
TELEMETRY	Temperature, dew point, humidity, pressure + trend, gas, **forecast**	—
CLOCK	Local time, date, sunrise and sunset	Zone · Format · Sync from GPS
LORA	Frequency, SF, BW, CR, power, counters, RSSI, S-meter, time on air	Next profile · Power · Reset counters · Heard list
SYSTEM	Heap, PSRAM, Flash, battery, uptime, Bluetooth and WiFi	—

3. Handy things

- **GO TO:** from LAST APRS, STATIONS (Browse mode) or an alarm: opens COMPASS pointing at that station. Beeps when you get within 50 m.
- **Quick reply** (in message detail): QSL, 73, TNX, WAIT, CALL ME, QSY CQ VHF, QSY CQ UHF, QSY RPT LOCAL. Sent requesting acknowledgement.
- **Beacon mode** (letter at the top right of MAIN): **A** automatic, **S** SmartBeaconing, **M** manual (only what you send with *Send beacon* goes out).
- **LoRa power:** LORA → LP → Power → +1 / −1 dBm. Saved and kept after reboot.
- **S-meter (VHF/UHF):** S9 = −93 dBm, 6 dB per point. Row `S8 -99 SNR 5.0` in LORA.

4. Emergencies

If you receive an emergency (Mic-E *Emergency*, or a message/comment with SOS, EMERGENCY, MAYDAY, MOB or MAN OVERBOARD): the screen turns on, the siren sounds and the LED blinks fast. Alarm screen with callsign, distance and bearing.

Press	Effect
1P	Silences (another 1P closes the screen)
LP	**GO TO**: opens COMPASS towards the station
2P	Silences and closes

If the emergency is you: 5 short presses or SOS in Morse (short = under 0.3 s; long = 0.3–0.7 s). 5 s countdown with beeps (**any press cancels it**). Then a Mic-E *Emergency* and a beacon with the text "EMERGENCY SOS" are sent, **every 30 s until cancelled**. Cancel by repeating the gesture or MAIN → LP → *Cancel SOS*. While it lasts, "SOS" blinks in the header and the GPS does not sleep.

5. MORSE mode

Single screen with the decoded text. **LP** opens the submenu:

Option	What it does
1–5	Sends the CW macro (editable in the WebUI, `{CALL}` = your callsign)
Listen CW (RF)	Decodes CW received over RF (carrier within ±2 kHz, ≥ 6 dB above noise)
Trainer: listen	5 Koch characters sound; you identify them by ear and they are revealed at 1.5 s
Trainer: keying	Key the shown character with your key; counts hits
CW beacon: ON/OFF	CW beacon every N min (WebUI): `DE {CALL}/B {LOC} V{VBAT}`
Clear text / Change mode	Clears the screen / returns to the selector

- The **button** aborts any send, listen or trainer.
- **Key:** 3.5 mm jack; tip = dit, ring = dah, sleeve = GND. Pins, type (straight, iambic A/B), WPM, sidetone and trainer level in the **WebUI** (LoRa → CW / Morse).
- During a send or listen the Tracker is paused (no beacons or APRS reception).
- Transmitting needs SX1262/SX1268. Sidetone and listen trainer need the buzzer enabled.

6. WSPR beacon mode

Single screen: callsign, GPS locator, frequency, UTC time and a **countdown** to the next transmission (110.6 s, starts 1 s after the even minute). **LP**: *Beacon ON/OFF*, **Tune carrier**, *Change mode*. The **button** aborts a transmission and leaves the beacon OFF.

- **WebUI:** centre frequency (70 cm: dial 432.300 MHz USB = `432301500` Hz; 433.600 is **not** WSPR), power (0–20 dBm, default 10), interval, optional fixed locator, offset.
- **Calibrate:** *Tune carrier* → measure with an SDR → `offset = nominal − measured`. ±100 Hz (±0.23 ppm) is required: the TCXO alone is not enough.
- GPS time is required. Use a 70 cm low-pass filter before the antenna. Amateur bands only.

7. If something is wrong

Symptom	What to check
"WAITING FOR GPS"	Open sky; in GPS → LP → *GPS on/off*. The position in MAIN comes from the GPS.
TELEMETRY says "Sensor: none"	Enable *Telemetry* in the web configuration and connect a BME280/BMP280/BME680.
Compass shows no heading	Keep the board flat; when stationary it uses the magnetometer, when moving the GPS course.
No alarm sounds	Check in the configuration that the buzzer and message LED are enabled.
Screen off	Any press turns it on.
WSPR: "NO GPS TIME"	Open sky until GPS time is available.
WSPR: nobody decodes me	Calibrate the frequency (offset) and check dial, filter and antenna.
MORSE / FSK missing in the selector	Only offered with SX1262/SX1268 (MORSE) and without LLCC68 (FSK).
"Trainer needs buzzer"	Enable the buzzer in the configuration.
"Set key GPIO (WebUI)"	Define the key pin in the WebUI.